# Ensemble of Hierarchical Temporal Memory for Anomaly Detection

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***Abstract*—Hierarchical Temporal Memory (HTM) is a continuously learning algorithm derived from neuroscience that models spatial and temporal streaming data. It was demonstrated that HTM produces a good performance in predicting unusual patterns or anomaly detection in univariate datasets. In this paper, we deploy the HTM algorithm for the anomaly detection problem in multivariate datasets, which are more common in practical scenarios. We first investigate the implementation of HTM using multi-encoders for multiple variables and analyze its performance in different parameter settings. Then, we introduce a new framework for ensemble learning by using single-encoder HTMs as weak learners. We carried out experiments on public datasets in different dimensions. Our experimental results show that our new approach outperforms the multi-encoder implementation of the HTM algorithm.**

***Keywords*-Anomaly Detection, Unsupervised Learning, Ensemble Learning, Hierarchical Temporal Memory.**

## I. Introduction

With the increasing growth of global Cybersecurity threats, developing an automated forewarning system is a severe challenge. Such a system must be able to filter out *anomalous* activities and work as a tool to prevent both *known* and *unknown* attacks. Traditional security tools, which are based on signatures, are not relevant. Although, they may be unknown, malicious attempts often differ from most normal activities and thus can be detected by identifying outlying data. Anomaly-based detection approaches are thus applicable in the context. This problem could be found in many applications, for example, fraud detection [1], computer network security [2], [3], medical and public health anomaly detection [4], [5], etc. Basically, an anomaly detection system first builds a "normal" profile for an user/system and then generates an anomaly alarm once there is a suspicious activity deviating sufficiently from the profiling. For example, to detect insider threats in network security, a security system first builds user profiles in the network, then monitors these users' daily activities (e.g., browsing web, downloading/uploading, logging on/off, using (network) resources, etc). Finally, any *malicious* activities, which significantly differ from the users' profiles, will be reported.

As the number of outliers is usually significantly smaller than the number of normal activities, identifying and excluding outliers, especially in real-time, is challenging for anomaly detection algorithm. Collecting and analyzing many data with many features may provide higher accurate detection; however, it costs the scalability of the detecting system. On the other hand, less data or a dataset with a few features may not produce accurate results.

Numerous machine learning models have been proposed for anomaly detection. Semi-supervised or unsupervised learning methods are often used as data is usually *unlabeled*, and the outliers represent new or unseen behaviours. These models for anomaly detection also avoid the bias introduced by supervised learning models with wrong labelled data. Furthermore, much of the world's data is streaming, time series data, anomaly detection techniques must be able to process in real-time and learn while simultaneously making predictions.

Hierarchical Temporal Memory (HTM) is a neuroscience-based intelligence model, described initially in [6] by Hawkins and Blakeslee. It captures the structural and algorithmic properties of the *neocortex*, which is in charge of visual pattern recognition and other cognitive processes [7]. Applying HTM as an unsupervised machine learning model to detect anomalies was reported in [8], [9]. Basically, this anomaly detection algorithm can be considered as a type of neural network with a much more solid biological foundation than other artificial neural networks (ANN). By modelling spatial and temporal patterns in data in an unsupervised fashion, this machine learning model meets real-time, continuous, online learning requirements without supervision.

### A. Contribution

HTM has demonstrated its efficiency in detecting anomalies with single feature datasets in [9]. To the best of our knowledge, there have not had any anomaly detection reports on multivariate datasets by using the HTM algorithm yet. However, it was stated that the algorithm works well with such datasets.

In this paper, we investigate how effective in applying the HTM algorithm in datasets with multiple features. Our contributions are summarized as follows:

- We deploy the HTM algorithm on publicly multivariate datasets, which are labelled and widely used for anomaly detection. To investigate how the dimension of datasets impacts the HTM algorithm's performance, we choose datasets with different numbers of features, the least has three features, while the most have 274 features. These datasets also have different ratios of outliers.

- We then introduce a new framework for ensemble learning via the HTM algorithm. In our framework, an HTM detector with a single feature works as a weak learner. Our idea is to design a *scalable* anomaly classifier that leverages HTM's advantages and is suitable for detecting anomalies in big data with multiple features. In this paper, we aim to explore our proposal in static datasets to evaluate the possibility of exploring HTM in this domain. Streaming settings are considered for our future work.
- Experimental results show that our approach outperforms the implementation of the HTM algorithm with multi-dimensional datasets.

The rest of the paper is organized as follows. Section II briefly recall some knowledge about anomaly detection and the HTM algorithm. We discuss the related works in Section III. Section IV describes our ensemble learning models. Section V details the public datasets used, evaluation metrics, and the experimental results. Finally, we conclude and provide some discussions in Section VI.

## II. Preliminaries

### A. Anomaly Detection

Anomaly Detection aims at finding data patterns that do not conform to expected behaviour. The problem has many real-world applications such as network/system security, medical diagnosis, online financial transactions, etc. Different aspects are considered while investigating the suitability of machine learning algorithms for anomaly detection based on user profiling. Few critical aspects include:

- **Nature of Input Data:** Data could be labelled or unlabeled. It is a key aspect of any anomaly detection problem to consider which type of machine learning algorithms could be applied.
- **Scalability:** Data is static or streaming? The algorithms must make a decision fast once receiving a new sample for handling streaming data because data stream samples arrive continuously and at high speed.
- **Output of Anomaly Detection:** An essential characteristic of an outlier algorithm is its output, which can be either scores or binary labels.
- **Data processing model:** There are two approaches: *Online processing* or *Online-Offline processing*. The former approach views the stream detection problem as a single-pass processing challenge. It utilizes a general adaptive method to update the detection model wh as new data objects arrive from the stream. It does not provide an investigation of the structure of the detection model at different time intervals. In contrast, the Online-Offline processing consists of two phases; an online step where the model prepares statistics over the streaming data in an online mode and an offline stage where the learned model would be performed upon these statistics regarding user-defined temporal predicates. This type of approach is more flexible in data stream exploration because it extracts multiple models from the streaming data by considering different time windows.

### B. Hierarchical Temporal Memory (HTM)

Hierarchical Temporal Memory (HTM) is a machine learning technique that is initially inspired by the human brain's general principles. It is an unsupervised learning algorithm based on the structural and algorithmic properties of the *Neocortex*. Although the Neocortex consists of many regions responsible for different tasks, the cellular structure is almost the same throughout all these regions. They share the same three characteristics; 1) hierarchical structure, i.e., signals flow in both directions of the hierarchy, 2) information stored in temporal in nature, and 3) a memory system trying to remember and predict patterns over time. This finding is crucial as this means that the brain solves similar problems to process any kind of sensory data, whether it is visual or audio, etc. The hierarchical structure of Neocortex leads to a hierarchical principle of the HTM algorithm. Fig. 1[1] depicts the hierarchical structure of HTM. The output of one level (region) will be passed to the following level (region) to be processed/trained, and the process continues for every region until the last one. This principle decreases memory usage and training time significantly because of reusing patterns learnt at each level of the hierarchy to combine them at the following level. The generalization of expected behaviour can be provided by sharing representations in a hierarchy that enables a new sample to inherit its sub-components' available properties.

HTM algorithm focuses on three main properties: *sequence learning*, *continual learning* and *sparse distributed representations*. Sequence learning refers to the ability to model temporally correlated patterns, and hence HTM can predict the future. Continual learning implies the ability to continue learning over time. This characteristic is essential to deal with stream data. Finally, sparse distribution representations mean that only a small portion of neurons are active at one moment and is not enough to realize something because several neurons should be activated for detection.

## III. Related Work

### A. Anomaly Detection Algorithms

Based on the availability of labels, anomaly detection techniques can be classified into three approaches; 1) *Supervised training models* assume that data are labelled as normal or abnormal; 2) *Semi-supervised training models* assume that training data has only labelled instances for the normal class, and 3) *Unsupervised training models* do not require any labelled data. Since real-world problems are complex with many uncertainties, supervised anomaly detection is not relevant. The unsupervised/semi-supervised anomaly detection systems are most widely applicable because they are designed to predict anomalies occurring from unknown situations.

In the literature, several algorithms have been proposed for anomaly detection. They are well studied in structured and comprehensive surveys [10], [11]. Based on the underlying

[1] https://www.taoeffect.com/other/nupic/

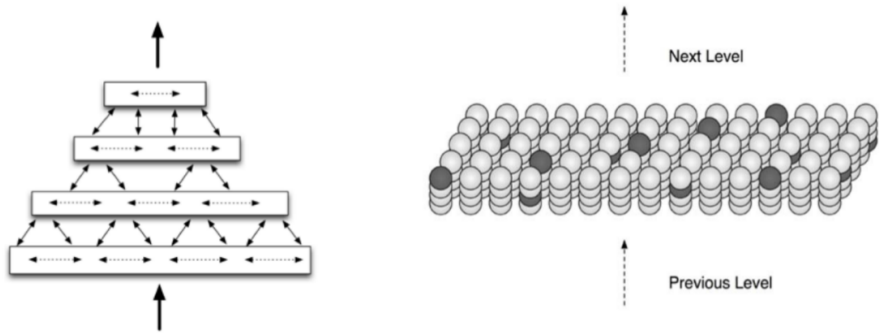



Fig. 1: Left figure shows the simplified diagram of four HTM regions arranged in four-level hierarchy. Right figure shows an HTM region with columns of cells.[2]

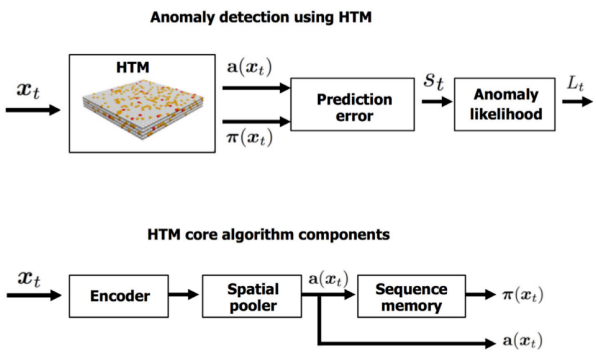



Fig. 2: Using HTM for Anomaly Detection [9].

technique adopted, anomaly detection methods can be categorized into the following groups:

- *Classification based methods*: the idea is to train a model with labelled data, and then the trained model will then be used to distinguish between normal and abnormal classes. One-Class SVM [12] is a typical example of this category.
- *Clustering based methods*: these methods assume that normal data belong to big clusters, while anomalies either belong to small clusters or do not belong to any clusters. The basic algorithms corresponding in this description include K-Means and DBSCAN [13].
- *Nearest neighbor based methods*: the assumption behind this category is that normal data appear in dense neighbourhoods, while anomalies are far from their closest neighbours. Local Outlier Factor (LOF) [14] is a well-known distance-based approach in this category.
- *Statistical methods*: the methods in this category assume that normal data occur in high probability regions of a stochastic model, while anomalies occur in the low probability regions of the stochastic model. They include Gaussian Mixture Model (GMM) [15] (parametric model), Kernel density estimators (KDE) [16] (non-parametric model).
- *Information theoretic models*: these methods assume that anomalies in data induce irregularities in the data set's information content. The Kullback-Leibler (KL) divergence was introduced in [17] as an information-theoretic method.
- *Ensemble learning methods*: they combine multiple underlying classifiers. This direction works on the idea that a combination of weak classifiers can produce a stronger classifier. Random Forest [18] is one of the most wide-used methods in this category.

### B. HTM for Anomaly Detection

Ahmad et al. [9] proposed a new anomaly detection algorithm based on Hierarchical Temporal Memory (HTM), an online algorithm built on sequential memory.

Fig. 2 shows how HTM algorithm works as anomaly detector. Given a real-time data stream $\ldots, x_{t-2}, x_{t-1}, x_t, x_{t+1}, x_{t+2}, \ldots$, where $x_t$ is a vector representing the state of a real-time system at time $t$, HTM models the temporal sequences in that stream. Input temporal data $x_t$ is first *semantically encoded* in a form of *sparse distributed representation* (SDR), then goes through a *spatial pooler* to be normalized/standardized into a sparse output vector or mini-columns of definitive size and fixed sparsity. The resulting vector $a(x_t)$, a sparse binary vector representing the current input, is then given to the core of the system, *sequence memory component*. This component models $a(x_t)$ and outputs a prediction in the form of another sparse vector $\pi(x_t)$, that represents the HTM network's internal prediction for $a(x_{t+1})$. The dimensionality of these two vectors is the same as the number of columns in the HTM network.

At the point $t+1$, we would like to determine whether $x_{t+1}$ is unusual by comparing actual values to the predictions made at the point $t$. An anomaly score between 0 and 1, depending on the "similarity" between the actual values and the predictions, will be determined. The SDR can be determined as the similarity measurement. So, the anomaly score inversely depends on the overlap between actual and predicted bits in column space. The anomaly score will be 0 if one of the predictions is exactly equal to $x_{t+1}$. Otherwise, the anomaly score will be 1 if any of the predictions are significantly different from $x_{t+1}$. Formally, the raw anomaly score is computed as follows:

$$s_{t+1} = 1 - \frac{a(x_{t+1}) \cdot \pi(x_t)}{|a(x_{t+1})|} \quad (1)$$

To handle the false positives causing by directly thresholding the raw anomaly score, HTM computes the *anomaly likelihood*. HTM models the distribution of anomaly scores and uses this distribution to determine how anomalous the current state is based on the prediction history. This history consists of a window of the last $W$ anomaly scores. By assuming the anomaly scores has a rolling normal distribution, then mean $mu_t$ and covariance $\sigma_t^2$ are continuously updated as follows:

$$\mu_t = \frac{\sum_{i=0}^{W-1} s_{t-i}}{W} \quad (2)$$

$$\sigma_t^2 = \frac{\sum_{i=0}^{W-1} (s_{t-i} - \mu_t)^2}{W-1} \quad (3)$$

In order to decide if an observation is anomalous, HTM model applies a threshold to the Gaussian tail probability:

$$L_t = 1 - Q\big(\frac{\tilde{\mu}_t - \mu_t}{\sigma_t}\big), \quad (4)$$

where $\tilde{\mu}_t = \frac{\sum_{i=0}^{W'-1} s_{t-i}}{W'}$ and $W' \ll W$ is a window for a short term moving average. The threshold $L_t$ is based on a user-defined parameter $\epsilon$. An anomaly is reported if $L_t \geq 1-\epsilon$.

HTM for anomaly detection was demonstrated its efficiency in [9]. In [19], authors also applied HTM to real-time and sequential problems with time-varying streaming input data. The obtained results denote that the model can continuously detect many variable order temporal sequences while utilizing an unsupervised Hebbian-like learning rule. Furthermore, the HTM algorithm is robust to noise, and it can learn various patterns simultaneously (high capacity). Since then, HTM has attracted the attention of many researchers to use this framework in different applications. For example, Wang et al. [20] proposed a distributed anomaly detection system using this framework to enhance the security of vehicular Controller Area Network (CAN) bus by predicting the flow data in real-time, which depends on the state of the previous learning. In [21], authors used the HTM technique to find performance anomalies on real-time infrastructure metrics, which are continuously collected by monitoring the resource consumption of executing workflow tasks. In [22], the authors report the results of the anomaly detection model in the health assessment as applied practice. Their HTM-based anomaly detection model learnt how to predict the diesel generator set and detect anomalies on the data collected from Wireless Sensor Network.

### C. Ensemble Learning

The main advantage of ensemble learning methods is that they can achieve a better performance than the individual techniques by combining multiple classifiers. Besides, ensemble methods can scale for large datasets. This section briefly reviews some ensemble learning methods applicable to network security issues.

In [23], the authors focus on the application of ensemble techniques to network security and anomaly detection. Different ensemble learning methods are investigated to enhance the detection of attacks and anomalies in network measurements, following a Super Learner as a particularly promising model.

Das et. al. proposed a active ensemble learning algorithm [24] in which tree-based models such as Isolation Forest (IF) [25], HS-trees (HST) [26], and RS-Forest (RSF) [27] are used as ensemble detectors. The purpose of this method is to find anomalies both for batch and streaming settings. Also, a novel formalism is introduced to describe the discovered anomalies to enhance the diversity of instances presented to the analyst without loss in the anomaly discovery rate. Furthermore, they suggested a novel approach for drift detection, which is robust for detecting them in data streams. The authors assumed an analyst's availability to provide the actual label for any sample in an interactive loop.

Pevny [28] proposed an ensemble learning method for anomaly detection named Lightweight Online Detector of Anomalies (LODA). This algorithm is suitable for stream and big data due to its low time and space complexity while obtaining suggested accurate results. It consists of a set of histograms that are denoted as weak classifiers and hold $b$ bins. Each histogram approximates the probability density of input samples, which are mapped into a single vector. LODA assigns each sample a score by computing the average of the logarithm of probabilities estimated on each projection vector. LODA's output is comparable to the sample's negative log-likelihood, which means that the less likely a sample is, the higher the anomaly value it gains.

Zhao et al. [29] introduced the LSCP algorithm to address two challenges, namely lack of ground truth and lack of a reliable way for selecting proper base detectors. Their framework defines a local region around a test sample using the agreement of it's nearest neighbours in randomly selected feature sub-spaces. Also, selecting combine base detectors with emphasis on data locality tries to explain the current ensemble method's limitations. Several variants of LSCP were proposed in [29], which keep the promising detectors for a second phase combination instead of using the single most competent detector.

## IV. Ensembles of Hierarchical Temporal Memory

Our framework is described as in Fig. 3. It consists of three phases: the first phase, so-called Bootstrap samples and selected features, Weak Learners, Ensemble Model.

### A. Bootstrap Samples and Selected Features

In this step, we use bootstrapping as a statistical technique to generate train samples for each weak learner. For each weak learner, it selects a randomly drawn training subset of the size of $M$ from an initial dataset of size $N$. These training subsets have enough statistical properties under some assumptions. These subsets can be considered as $i.i.d$ samples of the true

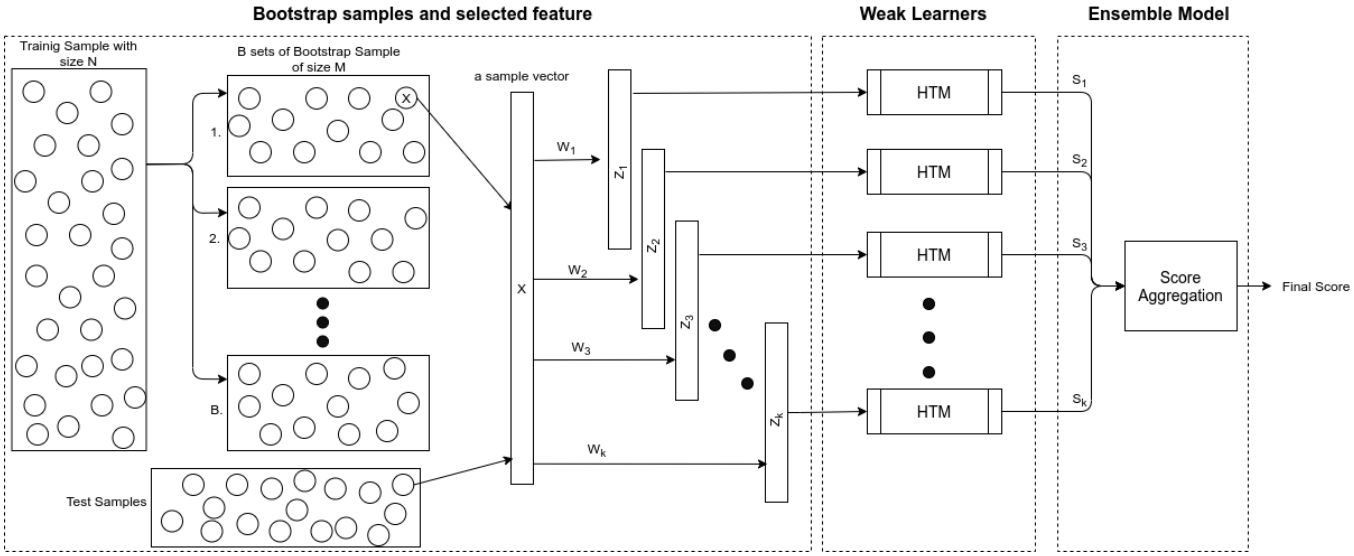



Fig. 3: Ensembles of Hierarchical Temporal Memory Framework

data distribution, so they are representative and independent samples of the true training data distribution.

After bootstrapping, the training set is comprised of a collection of $B$ subsets. Then, a set of sparse projection vector $\{w_i\}_{i=1}^{k}$ is defined and each sample projected into $k$ different one-dimensional space. These projection vectors are a key element in our proposed ensemble systems to enhance a single classifier's performance. They are initialized by randomly selecting at least $\sqrt{d}$ non-zero features, where $d$ is the dimension of the input samples, and generating at random non-zero items according to $N(0,1)$. We consider $\sqrt{d}$ non-zero features because it was demonstrated in [30] that the $L_2$ distance can be preserved with projected samples while selecting $\sqrt{d}$ non-zero elements. In addition, the consideration of the normal distribution $N(0,1)$ for non-zero items is because of the justification proven in [31]. The $L_2$ distances between samples in the projected space and the $L_2$ distances in the input space is approximately the same. The use of sparse random projection plays an essential role in increasing diversity among weak learners by feeding them different sub-spaces of the input space. It is a useful diversification technique which is widely used in well-known methods such as random forest [18].

In summary, our proposed method performs two kinds of sampling:

1) Sampling over the input data in the datasets to generate a bootstrap sample;
2) Sampling over features to keep only a random subset of features to feed data into weak learners.

The advantage of sampling over the features is that weak learners focus on different subspaces of problem space to make their decision, reducing the correlation between their outputs. Moreover, sampling over the features makes our proposed model's decision-making process more *robust* to missing data. The reason is that input data with the missing value may be classified by the weak learners whose projection vector $w_i$ doesn't contain the features having missing values.

### B. Weak Learner

Selecting the base models to be aggregated is an important step in designing an ensemble learning. Generally, there are two types of ensemble models: homogeneous and heterogeneous ensemble models. The former model consists of weak learners of the same type trained with different data. The later uses a set of weak learners of different types trained with the same data. In this paper, we design a homogeneous ensemble model that consists of $k$ number of HTM detectors as weak learners. Each HTM detector, $HTM_j$ for $1 \leq j \leq k$, is trained on a subsample of input data, $Z_j = \{x_i w_j^T\}_{i=1}^{M}$, in which each sample $x_i$ is projected onto one-dimensional latent space by using the vector $w_j$. Fig. 4 shows how a sequential set of samples of size ten is mapped into one-dimensional latent space using random projection vectors. In this example, there are two projection vectors, $w_1$ and $w_2$, and the input data $X$ mapped into two subspaces, $Z_1$ and $Z_2$. According to $Z_1$ subspace, $x_{t=7}$ and $x_{t=10}$ are outliers, and similarly $x_{t=7}$ is anomaly sample in $Z_2$ subsapce.

In our framework, several instances of one-dimensional HTM can be trained in parallel and independently from each other on different bootstrap samples with different subsets of features.

### C. Ensemble Model

An ensemble method tries to combine several learners into one predictive model to achieve better performance by reducing bias and/or weak learners' variance. One way to reduce the variance of the weak learners' outputs is to average them. So, given the outlier scores from HTM detectors $s_1, s_2, \ldots s_k$, the final step of our proposed ensemble model is to combine these scores by two commonly used functions:

- Averaging (avg): the average of scores predicted by HTMs is computed as the final score. Considering Eq.

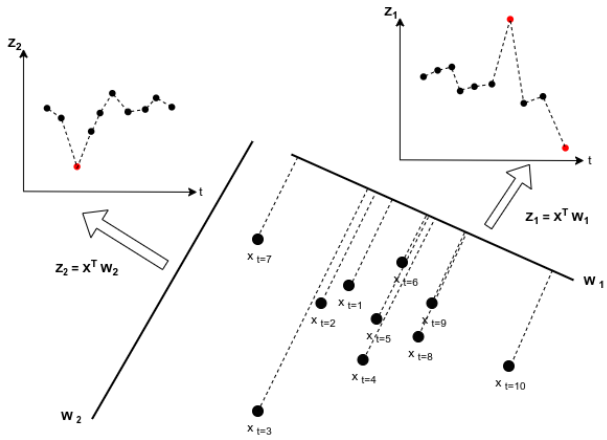



Fig. 4: How a sequential set of samples of size ten is projected onto two independent one-dimensional sub-spaces, $Z_1$ and $Z_2$, using two different random projection vectors $w_1$ and $w_2$.

4, the final score is defined as follows:

$$Score_{avg}(x_t) = \frac{1}{k}\sum_{j=1}^{k} s_j = \frac{1}{k}\sum_{j=1}^{k} 1 - Q(\frac{\mu_{t_j} - \mu_{t_j}}{\sigma_{t_j}}) \tag{5}$$

- Maximum (max): the maximum score across all HTMs return scores is considered as the final score.

$$Score_{max}(x_t) = \max_{j=1}^{k}(s_j) = \max_{j=1}^{k}(1 - Q(\frac{\mu_{t_j} - \mu_{t_j}}{\sigma_{t_j}})) \tag{6}$$

- Median (med): the median score across all HTMs return scores is considered as the final score.

The learning routines of our proposed ensembles of HTM are summarized in Algorithm 1. For the sake of simplicity, Algorithm 1 applies the averaging model when calculating scores. However, it is straightforward to apply other ensemble models, i.e., Maximum and Median.

**Algorithm 1:** Learning Ensembles of HTM (average).

**input :** input data samples $D = \{x_n \in R^d\}_{n=1}^N$
**output:** scores of input data samples $\{s(x_n) \in R\}_{n=1}^N$

initialize projection vectors with at least $\sqrt{d}$ non-zero elements $\{w_j\}_{j=1}^k$

**for** $t = 1$ *to* $B$ **do**
  $D_t = \{$ bootstrap $M$ samples from $D\}$
  $S_t = \{\}$
  **for** *each sample* $(x) \in D_t$ **do**
    **for** $j = 1$ *to* $k$ **do**
      $z = xw_j^T$
      $s_j = HTM_j(z)$
    **end**
    $S_t = S_t \cup \{\frac{1}{k}\sum_{j=1}^{k}(s_j)\}$
  **end**
**end**

**for** $n = 1$ *to* $N$ **do**
  $s(x_n) =$ avg $(\{s_i(x_n) | s_i(x_n) \in S_i \wedge x_n \in D_i\}_{i=1}^B)$
**end**

**return** $\{s(x_n)\}_{n=1}^N$

## V. Experimental Results

To conduct our anomaly detection experiment, we select 12 well-known benchmarks and evaluate the performance of our ensemble method and the most known algorithms such as LOF [14], CBLOF [32], One class SVM [12], LODA [28], and HTM [9]. Table I summarizes the characteristics of selected real-world outlier detection benchmark datasets from UCI [33], ODDS[2], and DAMI[3]. As it is clear in Table I, a wide spectrum of data sets is selected in order to examine the proposed method. For instance, *smtp* is a low dimensional data set while *arrhythmia* is a high dimensional one. *Wine* is a small data set with a low number of samples; in contrast, *smtp* is a large data set with 95,156 instances. The outlier rate varies from $0.03\%$ in *smtp*, which is very *imbalanced*, to $36\%$ in *ionosphere*. Glass is a multi-class dataset that was inspired by the criminological investigation. Each attribute in this dataset is real value, and no missing values exist. The shuttle is also a multi-classification dataset, and its attributes are integer values. The purpose of annthyroid dataset is to determine whether a patient referred to the clinic is hypothyroid or not. Attributes of the datasets are real values. Satellite dataset attributes are also integer values. Pima is a binary classification dataset in which patients have diabetes or not. Wine is a multi-classification dataset in UCI where one class is considered as an anomaly. This dataset has 13 attributes that are real and integer values. Thyroid dataset is similar to annthyroid with the same goal to find hypothyroid patients. The difference is that it is a multi-classification dataset that two classes are normal. Ionosphere dataset is composed of radar data in two classes, either good or bad. Its attributes are real and integer values. Breastw is a classification dataset, which records the measurements for breast cancer cases. There

TABLE I: Summary of data sets used in experiments

| Dataset | #samples | #features | #outliers |
|---|---|---|---|
| shuttle | 49097 | 9 | 3511 (7 %) |
| annthyroid | 6832 | 6 | 534 (7.42 %) |
| satellite | 6435 | 36 | 2036 (32 %) |
| pima | 768 | 8 | 268 (35 %) |
| wine | 129 | 13 | 10 (7.7 %) |
| thyroid | 3772 | 6 | 93(2.5 %) |
| ionosphere | 351 | 32 | 126 (36 %) |
| breastw | 683 | 9 | 239 (35 %) |
| arrhythmia | 452 | 274 | 66 (15 %) |
| glass | 214 | 9 | 9 (4.2 %) |
| mammography | 11183 | 6 | 260 (2.32 %) |
| smtp | 95156 | 3 | 30 (0.03 %) |

[2]http://odds.cs.stonybrook.edu/
[3]https://www.dbs.ifi.lmu.de/research/outlier-evaluation/DAMI/

TABLE II: ROC-AUC of different anomaly detection methods using 12 well-known benchmarks.

| Dataset | LOF | CBLOF | OCSVM | LODA | Iforest | HTM | Ensemble-HTM (avg) | Ensemble-HTM (max) | Ensemble-HTM (median) |
|---|---|---|---|---|---|---|---|---|---|
| shuttle | 55.51 | 78.97 | 72.49 | 98.75 | 99.76 | 10.27 | 82.19 | 82.30 | 70.99 |
| annthyroid | 73.92 | 50.71 | 54.67 | 70.54 | 81.16 | 51.36 | 54.85 | 55.45 | 50.57 |
| satellite | 53.96 | 56.29 | 48.73 | 72.02 | 66 | 55.06 | 67.30 | 64.73 | 64.77 |
| pima | 53.84 | 70.29 | 53.93 | 61.02 | 67.39 | 54.31 | 56.88 | 54.86 | 57.5 |
| wine | 82.48 | 88.15 | 47.52 | 91.06 | 77.48 | 54.2 | 87.89 | 74.07 | 89.07 |
| thyroid | 80.24 | 86.55 | 84.37 | 92.8 | 97.87 | 75.14 | 71.15 | 70.71 | 57.43 |
| ionosphere | 87.18 | 79.54 | 76.21 | 83.43 | 86.61 | 40.85 | 69.42 | 68.19 | 57.36 |
| breastw | 37.93 | 99.49 | 97.87 | 98.52 | 98.78 | 54.63 | 86.25 | 77.93 | 86.59 |
| arrhythmia | 78.85 | 81.02 | 46.79 | 78.93 | 80.23 | 48.77 | 63.22 | 63.9 | 60.41 |
| glass | 83.25 | 80.16 | 59.35 | 81.25 | 72.9 | 25.06 | 67.47 | 65.79 | 61.65 |
| mammography | 71.93 | 84.49 | 87.21 | 83.91 | 86.84 | 64.48 | 61.97 | 61.59 | 59.57 |
| smtp | 90.33 | 80.51 | 84.15 | 93.31 | 89.8 | 79.03 | 82.76 | 81.49 | 81.01 |

are two classes, benign and malignant with integer valued attributes. Arrythmia dataset is used to distinguish between the presence and absence of cardiac arrhythmia. Its attributes values are integer and real. Mammography dataset is a binary classification dataset combined of a non-calcification class as inlier, and calcification is the outlier. SMTP is a subset of the KDDcup dataset in which only SMTP service data is used.

In our implementations of LOF and CBLOF methods, the number of nearest neighbours and clusters was set to 10 and 8, respectively. We implemented OCSVM with a RBF kernel, and $\nu = 0.05$. Our implementation of the IForest used parameters as recommended in [25]: the number of trees was set to 100 and 256 randomly selected samples were used to construct a tree in the Iforest. The number of histograms in Loda, $k$, and the number of projection vectors in our proposed method $k$ optimized automatically as recommended in [28]. The hyperparameter setting of HTM followed [9].

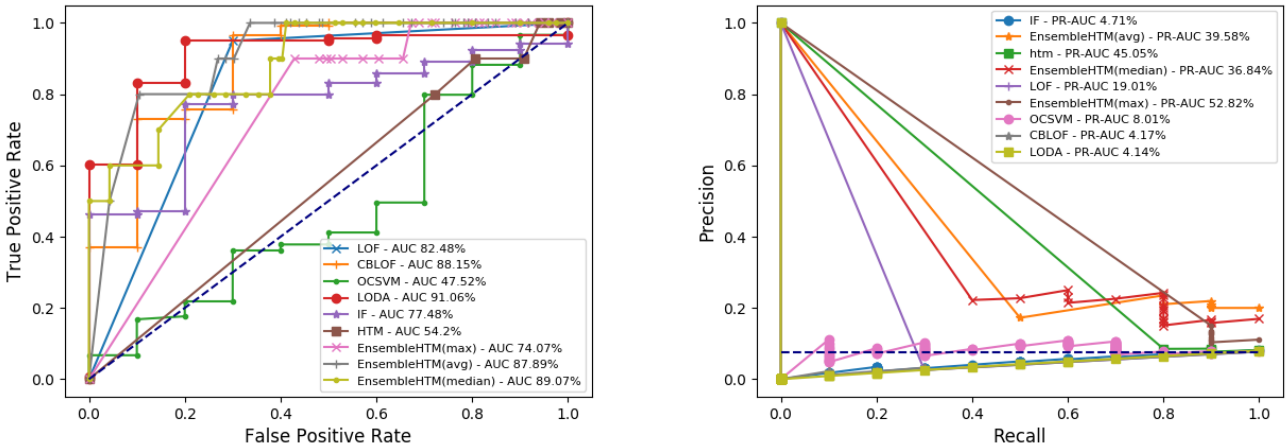
For reliable comparison, we evaluate the performance of models by taking the average score of ten independent trials using well-known metrics, including *precision*, *recall*, and area under the receiver operating characteristic curve (ROC-AUC). A ROC curve plots true positive rate (TPR) against false positive rate (FPR) at different thresholds. ROC-AUC measures the entire two-dimensional area underneath the entire ROC curve. So, it provides an aggregate measure of a detector's performance across all possible thresholds. AUC is scale-invariant, which means measuring how well scores are ranked, rather than their absolute values. Table II shows the ROC-AUC of different variants of our proposed method, along with other anomaly detection methods. Our proposed method performs better than HTM in terms of ROC-AUC. But, its ROC-AUC is not better than other learners such as Iforest in some datasets.

In many outlier detection applications, datasets are highly *imbalanced*, where the number of outliers is significantly smaller than the number of inliers. So, the outlier detection problem can be considered as a binary class imbalance problem where the anomaly class is under-represented compared with the normal class. Therefore, ROC-AUC cannot reflect classifiers' performance in outlier detection problems because it shows how the number of correctly labelled normal samples varies with the number of outlier examples classified incorrectly as normal data. Here, we have a problem of imbalance learning as the number of outliers in the datasets are varying 0.03% to 36% of the total sample. Therefore, we care more about the proposed method's performance in identifying the outliers samples than in identifying the inliers ones. The precision and recall of detecting outliers are better metrics to measure detectors' performance while using imbalance datasets. Precision is not affected by the issue of imbalanced data because it compares false positives to true positives rather than true negatives.

Tables III and IV respectively show the precision and recall of our proposed method along with state-of-the-art outlier detection methods. The results indicate that these approaches are likely to predict all samples as inlier, leading to higher AUC. But, in terms of outlier detection, they are less effective than our proposed method. Our proposed ensemble method can achieve better performance in detecting outliers than the state-of-the-art algorithms. The average version generally achieves better results in all datasets since it pays equal attention to all weak learners. Also, the median approach depends on the different range of values. Therefore, the median and average results are almost the same. But, their difference in some datasets is where the scale of weak learners' scores are scattered. On the other hand, the Maximum variant's performance varies on different datasets because of greedy behaviour for selecting the highest score of one weak learner.

Fig. 5 illustrates both ROC and Precision-Recall curve of different detectors using wine dataset. The Precision-Recall curve is better to calibrate the probability threshold in an imbalanced class scenario as a ROC curve. It summarises the predictive model's performance by considering the trade-off between the positive predictive value and the true positive rate using different probability thresholds.

We use Friedman's test [34] as a non-parametric multiple-comparison test to find significant differences across multiple methods. This test's null hypothesis is "*whether all algorithms have the same performance on average*?". The computed Friedman statistic for the ROC-AUC of all detectors considering all datasets is $44.57$. As this value is greater than the *critical value* at the significance level of $\alpha = 0.05$ (that is,




(a) ROC curve.

(b) Precision recall curve.

Fig. 5: The ROC and PR curve of different detectors using wine dataset.

TABLE III: Precision of detecting outliers of different methods using 12 well-known benchmarks for outlier detection.

| Dataset | LOF | CBLOF | OCSVM | LODA | Iforest | HTM | Ensemble-HTM (avg) | Ensemble-HTM (max) | Ensemble-HTM (median) |
|---|---|---|---|---|---|---|---|---|---|
| shuttle | 6.96 | 2.51 | 3.61 | 0.37 | 0.28 | 8.02 | 27.04 | 44.36 | 33.8 |
| annthyroid | 3.3 | 6.96 | 6.13 | 3.58 | 2.08 | 7 | 12.98 | 16.68 | 17.55 |
| satellite | 31.09 | 24.57 | 32.79 | 22.88 | 28.66 | 30 | 47.49 | 37.44 | 47.26 |
| pima | 31.34 | 22.73 | 33.89 | 29.96 | 24.29 | 25 | 40.93 | 38.71 | 41.37 |
| wine | 1.18 | 1.11 | 8.45 | 1.27 | 2.86 | 3.45 | 31.36 | 20.21 | 33.23 |
| thyroid | 0.75 | 0.46 | 0.53 | 0.25 | 0.11 | 6.02 | 11.14 | 9.16 | 8.05 |
| ionosphere | 13.67 | 15.1 | 21.29 | 14.39 | 13.22 | 17.09 | 45.27 | 46.04 | 40.91 |
| breastw | 45.3 | 10.88 | 11.8 | 11.54 | 11.33 | 37.48 | 57.26 | 54 | 65.31 |
| arrhythmia | 6.14 | 5.78 | 16.24 | 6.19 | 6.05 | 14.32 | 23.14 | 28.02 | 22.29 |
| glass | 1.39 | 1.43 | 3.61 | 1.4 | 1.92 | 1.01 | 4.43 | 5.42 | 8.24 |
| mammography | 1.38 | 0.69 | 0.52 | 0.7 | 0.58 | 2 | 7.22 | 8.43 | 12.22 |
| smtp | 0.01 | 0.01 | 0.01 | 0.01 | 0.01 | 0.2 | 0.32 | 0.54 | 1.09 |

15.51), we can reject the null hypothesis that all the learners perform the same on average.

We then perform a post-hoc test to examine differences in the algorithms' performance further. We applied the post-hoc Nemenyi test [35] with a significance level of $\alpha = 0.05$ to compare all detectors with each other. This test is based on the critical distance (CD) among methods, showing the minimum distance between two detectors to consider that they are statistically different. Fig. 6 shows the comparison of all methods' performance in terms of ROC-AUC, precision and recall against each other with the Nemenyi test. If the distance between the ranks of the two detectors' performance is larger than the critical distance (CD), we can consider that they have different performances. As Fig. 6a shows, the Nemenyi test, at $\alpha = 0.05$ could not find significant differences among the performance of the proposed method and the other learners in terms of ROC-AUC. However, the performance of Ensemble-HTM(avg) and Ensemble-HTM(max) in terms of precision and recall are different from the performance of LODA, CBLOF, and Iforest methods (see Fig. 6b).

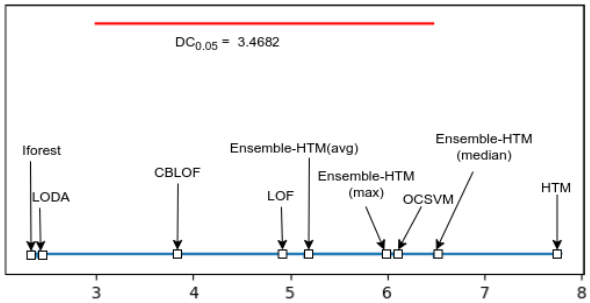
## VI. Conclusions and Discussions

In this paper, we investigated how the HTM algorithm is deployed in the multi-dimensional datasets. We proposed an ensemble of HTM detectors that leverage two effective sampling techniques, namely sampling over the input data and sampling over features to generate, to increase diversity among the output of weak learners. We carried out and reported experiments on 12 well-known publicly datasets with different dimensions, varying from a few dimensions to a few hundred dimensions. Our experimental results showed that an ensemble learning using the HTM algorithm provides better results than its implementations with multi-encoders. Moreover, our proposed ensemble learning has obtained comparable ROC-AUC in comparison with state-of-the-art outlier detection methods. On the other hand, it can achieve better performance in terms of precision and recall of detecting outliers than the state of art algorithms.

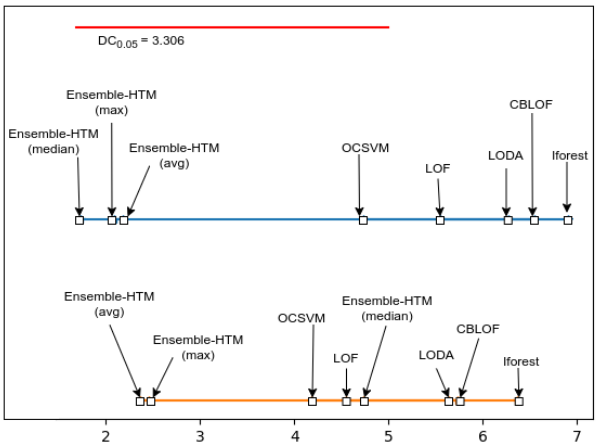
As future work, we would like to develop a new sketch for implementing our proposed method in a streaming setting and perform large-scale experiments with real data.

TABLE IV: Recall of detecting outliers of different methods using 12 well-known benchmarks for outlier detection.

| Dataset | LOF | CBLOF | OCSVM | LODA | Iforest | HTM | Ensemble-HTM (avg) | Ensemble-HTM (max) | Ensemble-HTM (median) |
|---|---|---|---|---|---|---|---|---|---|
| shuttle | 44.97 | 23.11 | 28.77 | 4.92 | 3.79 | 45.02 | 67.7 | 53.53 | 23.85 |
| annthyroid | 28.31 | 49.3 | 45.79 | 31.49 | 21.58 | 17.01 | 13.09 | 23.1 | 10.9 |
| satellite | 47.27 | 45.7 | 50.83 | 34.95 | 39.07 | 47.31 | 61.37 | 82.3 | 48.48 |
| pima | 47.61 | 36.81 | 47.29 | 42.83 | 38.61 | 79.32 | 54.13 | 52.55 | 54.53 |
| wine | 4.68 | 4.31 | 51.04 | 6.02 | 19.34 | 75.51 | 61.73 | 82.72 | 55.95 |
| thyroid | 20.95 | 14.71 | 16.75 | 8.41 | 3.38 | 63.48 | 31.9 | 23.38 | 7.49 |
| ionosphere | 26.34 | 31.12 | 33.3 | 28.76 | 26.62 | 73.08 | 59.27 | 57.66 | 31.38 |
| breastw | 61.79 | 26.51 | 28.58 | 28.02 | 27.49 | 37.49 | 86.87 | 67.39 | 77.9 |
| arrhythmia | 25.42 | 23.57 | 22.64 | 25.35 | 24.24 | 78.89 | 46.43 | 49.2 | 17.06 |
| glass | 17.52 | 19.07 | 39.95 | 18.78 | 26.77 | 26.02 | 25.88 | 30.76 | 61.26 |
| mammography | 35.15 | 22.66 | 17.41 | 23.49 | 19.58 | 85.1 | 30.9 | 22.81 | 15.85 |
| smtp | 11.03 | 21.83 | 17.4 | 10.99 | 15.53 | 56.31 | 3.5 | 3.33 | 3.52 |




(a) ROC-AUC.




(b) Precision(above) and Recall(below).

Fig. 6: Comparison of the total ROC-AUC, precision and recall of outlier detection of all detectors against each other with the Nemenyi test.

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